

Draft version

Two robust approaches for classifying time series from dynamical systems

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Keywords: time series; classification; dynamical systems; NeuralODE; inertial measurement unit

The task of time series classification is to assign class label to given time series. The number of classes is fixed. It is assumed that each class corresponds to some dynamical system. Each system generates phase trajectories within some probabilistic model. These trajectories are then mapped into empirical observations. The paper investigates how to restore dynamical systems out of experimental data using NeuralODE. We consider fixed and bayesian approaches to defining dynamical systems' parameters. Two classification principles are introduced: bayesian testing and transition to dynamical system's parameters space. Taken's theorem is used to approximate true

phase trajectories. At last, time series from inertial measurement unit are classified using given techniques and compared with other machine learning models.

1 Introduction

The paper introduces and analyzes methods for time series classification. The *time series* is a sequence of scalar values $x(t_i) = \{x(t_1), \dots, x(t_n)\}$ defined in discrete times $t_1, \dots, t_n$. The *multivariate* time series is a sequence of vector values $\mathbf{x}(t_i) = \{\mathbf{x}(t_1), \dots, \mathbf{x}(t_n)\}$ where values' dimension is $\mathbb{R}^s$. This is equal to having s time series on the same time domain. Everywhere further we assume time series to be multivariate in general. On practice we observe time series $\tilde{\mathbf{x}}(t_i)$ distorted by noise $\epsilon(t_i)$ so $\tilde{\mathbf{x}}(t_i) = \mathbf{x}(t_i) + \epsilon(t_i)$. We assume the $\epsilon(t_i)$ to be a zero-mean discrete *stochastic process*. Next, any time series has an associated *class label* $y \in 1 \dots K$. Here K is the number of classes. A *classifier* is a function $\delta(\tilde{\mathbf{x}}(t_i))$ that associates any time series with a class label. To measure classification quality we introduce a *risk function* $r(\delta)$. Now, formulate time series classification problem as:

Problem 1. *Given a set of N labeled time series $\mathcal{D} = \{(\tilde{\mathbf{x}}_1(t_i), y_1), \dots, (\tilde{\mathbf{x}}_N(t_i), y_N)\}$ solve an optimization problem $r(\delta) \rightarrow \min$. The noise terms $\{\epsilon_1(t_i), \dots, \epsilon_N(t_i)\}$ are assumed to be mutually independent processes.*

There are many methods for time series classification in the literature. *Distance-based* approaches rely on measuring similarity between time series using distance metrics. The metric can be defined using Dynamic Time Warping [1] and the k-Nearest-Neighbors (k-NN) classifier can be used then [2]. *Feature-based* approaches extract finite set of statistical [3, 4] or spectral [5] features from time series for further classification. *Deep learning* approaches uses neural networks to automatically learn time series features from raw train data. Convolutional neural networks [6, 7], recurrent neural networks (RNN) [8] or their combination [9–11] are designed to process sequential data like time series.

To solve the classification problem 1 we need to clarify the rule of assigning class labels to the time series. We also need to introduce a model of the time series. In the paper we assume this model to be a *dynamical system*. We define it as a time-invariant *ordinary differential equation (ODE)* of the form

$$\begin{cases} \frac{d}{dt}\mathbf{z}(t) = f(\mathbf{z}), \\ \mathbf{z}(0) = \mathbf{z}_0, \end{cases} \quad (1)$$

where $\mathbf{z} \in M$, $f(\mathbf{z}) : M \rightarrow M$ is a *vector field*, $\mathbf{z}_0$ is a *starting point*, M is an m -dimensional differentiable manifold. To find an $\mathbf{z}(t)$ satisfying (1) on some time interval is to solve an *initial value problem*. Under some regularity conditions on f the solution is unique [12]. The $\mathbf{z}(t)$ is called *phase trajectory*. To connect the dynamical system with the time series we introduce an *observation map* $\alpha : M \rightarrow \mathbb{R}^s$. We say a time series $\mathbf{x}(t_i)$ is generated by the phase trajectory $\mathbf{z}(t)$ if $\mathbf{x}(t_i) = \alpha(\mathbf{z}(t_i))$ for all t_i . Further we assume all time series to be generated. Next, consider K different dynamical systems with the vector fields f_j , $j \in 1, \dots, K$. If a time series is generated by a phase trajectory from the system j , the time series' label is $y = j$. Hence, we have completely described the time series model. Note that $\mathbf{z}(t_i)$ are not observed and therefore are not part of the $\mathcal{D}$.

Our time series' model can be applied to many real-word classification problems. For example, in medicine an electroencephalogram (EEG) is a direct, macroscopic measure of the brain's electrical activity [13]. The activity is governed by the Maxwell's Equations that can be reduced to the ODEs [14]. The EEG can be used for detecting arrhythmias or other heart conditions [6] as well as to identify seizures and sleep stages from brainwave data [7]. In the weather forecasting the fundamental equations of atmospheric dynamics can also be reduced to an ODEs system [15]. The classification problems in this field include weather regime classification [16] and climate pattern

identification [17]. In human activity recognition (HAR) ODEs are primarily used to model the dynamics of human motion. The core idea is that human movement is a continuous physical process governed by laws of mechanics, which can be described by differential equations [18]. This idea can be applied to the basic or fine-grained activity recognition [19, 20], fall detection [21], and other tasks. To conclude, our model is applicable when it is reasonable to assume that observed time series come from some dynamical system of the form (1).

To solve the problem 1 we first *reconstruct* the phase trajectories $\tilde{\mathbf{z}}(t_i)$ from the observations $\tilde{\mathbf{x}}(t_i)$. The $\tilde{\mathbf{z}}(t_i)$ will be the noisy estimates of the $\mathbf{z}(t_i)$. The reconstruction will be based on building an embedding of the manifold M . In general, an embedding of a manifold M is a smooth map $\phi : M \rightarrow N$ into another manifold N that is a diffeomorphism onto its image $\phi(M)$. It follows that phase trajectories $\mathbf{z}(t) \subset M$ and $\phi(\mathbf{z}(t)) \subset \phi(M)$ are basically identical (more strictly, diffeomorphic). One approach to build an embedding from the time series is the Whitney Embedding Theorem [22]. If α is a generic map from M to $\mathbb{R}^{2m+1}$ then α itself is an embedding. Therefore, the time series $\tilde{\mathbf{x}}(t_i)$ are the reconstructed $\tilde{\mathbf{z}}(t_i)$. However, this approach requires an excessive $2m+1$ number of time series even when m is rather small. Another approach is the Takens embedding theorem [23]. It requires only one observed time series $x(t_i)$. If α is a generic map from M to $\mathbb{R}$ then the embedding is given by the map $\phi(\mathbf{z}) = (\alpha(\mathbf{z}), f^{-1}(\alpha(\mathbf{z})), \dots, f^{-k}(\alpha(\mathbf{z})))$ where $k > 2m$. If we apply ϕ to the $\mathbf{z}(t_i)$ we obtain $\phi(\mathbf{z}(t_i)) = (x(t_i), x(t_{i-1}), \dots, x(t_{i-k}))$ which we call *delay vector* and denote as $\overleftarrow{\mathbf{x}}(t_i)$. If we compose a delay vector from the noisy time series $\tilde{x}(t_i)$ then it will be the reconstructed $\tilde{\mathbf{z}}(t_i)$. Illustration of the Takens theorem is on the Fig. 1.

After reconstructing the phase trajectories we will build approximations $\hat{f}_j$ for the system's vector fields f_j . Our approach for the approximation is the Neural Ordinary Differential Equations (NODE) [24]. It allows to use gradient optimization when

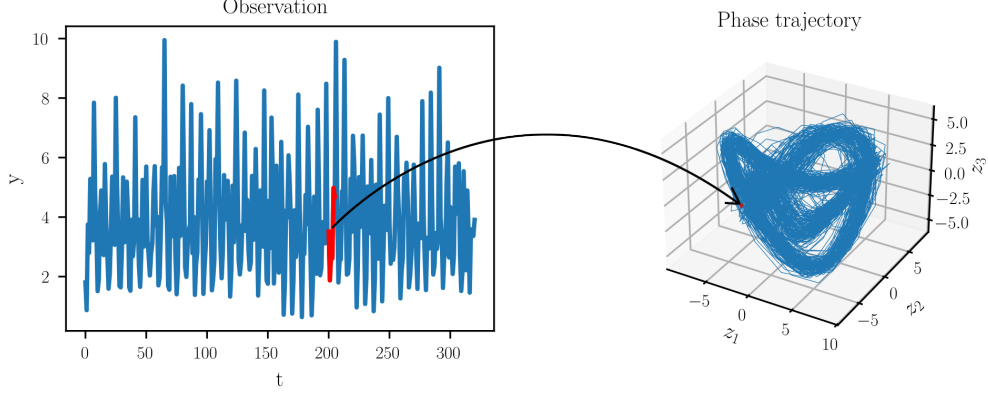


Figure 1 Application of the Takens theorem. On the left are observations $x(t)$. To build a point of the latent phase trajectory $\mathbf{z}(t)$ we must take several lagged observation values (colored in red), e.g. $\mathbf{z}(t_i) = (x(t_i), x(t_{i-1}), x(t_{i-2}))^T$.

an optimized objective depends on ODE trajectories. There are plenty of papers using NODE for regression tasks. For example, NODE are used in the context of the physics-informed ODE [25, 26] and control learning [27, 28]. However very few papers analyze NODE in the context of time series classification. Authors of [29] reinterpreted convolutional neural networks in terms of latent ODE dynamics and used it for hyperspectral image classification. Paper [30] solved classification problem in terms of control theory and NODE. Our paper contributes to the NODE applications for the general dynamical systems.

Actually, the NODE paper [24] introduced its own time series model called latent ODE. It is illustrated on the Fig. 2. The model is probabilistic and is described as

$$\begin{aligned} \mathbf{z}_{t_0} &\sim p_0(\mathbf{z}_{t_0}), \\ \mathbf{z}_{t_1}, \mathbf{z}_{t_2}, \dots, \mathbf{z}_{t_N} &= \text{ODESolve}(\mathbf{z}_{t_0}, f, t_0, \dots, t_N), \\ \text{each } \mathbf{x}_{t_i} &\sim p_x(x|\mathbf{z}_{t_i}) \end{aligned}$$

where $\text{ODESolve}()$ is the solution of the initial value problem. Authors then used a variational autoencoder (VAE) [31] to inference the distributions p_0, p_x and the vector field f . While this model is more general and is widely used [32–34], it has several drawbacks compared to ours. First, it has to parameterize f, p_x and a VAE. Hence, more data is needed in general to obtain good approximations of these functions. We only need to parameterize f . Second, the authors use VAE to obtain $\mathbf{z}(t_0)$ that does not guarantee good approximation [?]. We can reconstruct $\mathbf{z}(t_0)$ using the general results discussed above. Third, in the NODE model a connection between the phase trajectories and the observed time series is given by the general distribution p_x . Thus, the connection is difficult to interpret. In addition, the VAE is needed to reconstruct the $\mathbf{z}(t_i)$ from the $\mathbf{x}(t_i)$. Our reconstruction approaches give explicit and easy ways to obtain $\mathbf{z}(t_i)$ from the $\mathbf{x}(t_i)$ and vice versa. Several papers that use Takens theorem ideas with the NODE to restore latent dynamics can be found in the literature [35, 36]. But unfortunately, it is poorly covered for machine learning applications. Our paper aims to close this gap.

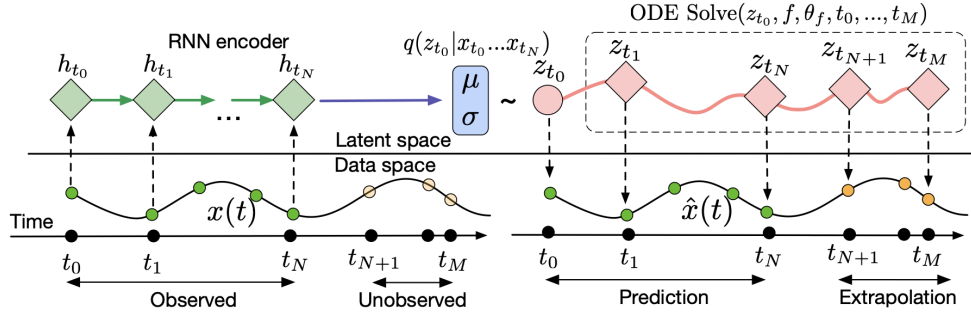


Figure 2 Computational graph of the latent ODE model. Source: reprinted from Chen et al.(2018)

To build a classifier we will first introduce two probabilistic models to connect class label and dynamical system. The models will be expressed as *probabilistic graphical models* [37]. Then, we will use bayessian classification [37], variational principle [?] or change of variables [38] as means for different classification methods. They are all

theoretical principles from statistics that lead to particular classifier. We will explore them later.

Every classification method is negatively effected by numerical approximations of solving (1). This is because NODE is based on imprecise numerical ODE solvers [39]. We will show how the classification quality decreases depending on the length n of the latent trajectory $\mathbf{z}(t_i)$. So limited trajectory length is an explicit constraint of our approach. Another reason for this limit is inaccurate vector field reconstruction from the labeled data. We will include it into the analysis as well. Finally, the latent dynamical system itself may be inadequate model for the whole time series. But it can fit time series locally. So we have limit on the trajectories again. Let's call the maximum trajectory length that can be properly classified as *forecasting horizon* and denote it as K . More formal definition will be given later in the paper. If we have trajectory to be classified longer than the forecasting horizon, the general solution we suggest is to split the trajectory on chunks of the length K . We will show how to adapt the classification methods for this case. Note that usually numerical approximation are not accounted in NODE applications but for us they are crucial and must be analyzed.

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